



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: Case Study Analysis

Case Study Title: Smart Waste Collection System

(SDG 11: Sustainable Cities and Communities)

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1. Introduction

Background of the Problem

Many municipalities collect garbage using fixed schedules without considering whether waste bins are full or empty. As a result, some bins overflow before collection, creating unhygienic conditions, while other bins are collected even when nearly empty. This leads to inefficient use of fuel, manpower, and time.

A Smart Waste Collection System uses IoT-enabled smart bins, GPS tracking, cloud storage, and route optimization software to monitor bin status in real time. The system helps municipal authorities improve waste collection efficiency and maintain cleaner cities.

Importance of Solving the Problem

- Prevents overflowing garbage bins
- Reduces fuel consumption
- Improves public hygiene
- Saves operational costs
- Supports smart city initiatives
- Protects the environment

Related SDG Goal

SDG 11 – Sustainable Cities and Communities

Targets:

- Improve urban sanitation
- Promote sustainable waste management
- Build cleaner and healthier cities

Objectives of the Case Study

- Study the existing waste collection process.
- Identify current challenges.
- Design a smart waste management software solution.
- Improve route planning.
- Increase operational efficiency.
- Support SDG 11.

2. Problem Understanding

2.1 Problem Description

Currently, garbage trucks visit every waste bin according to a fixed schedule regardless of whether the bin is full or empty. Overflowing bins create hygiene problems and unpleasant

surroundings. Empty-bin collection wastes fuel and manpower. Municipal authorities need a software solution that monitors waste levels, provides real-time alerts, and generates optimized collection routes.

2.2 Objectives

- Reduce manual monitoring
- Prevent waste overflow
- Optimize collection routes
- Reduce fuel consumption
- Improve operational efficiency
- Monitor bins in real time
- Generate reports automatically
- Improve citizen satisfaction

2.3 Scope

In Scope

- Smart bin monitoring
- GPS tracking
- Route optimization
- Driver management
- Admin dashboard
- Notifications
- Report generation
- Waste collection history

Out of Scope

- Manufacturing smart bins
- Waste recycling operations
- Vehicle maintenance management
- City tax collection

3. Stakeholder Analysis

Stakeholder	Role	Responsibilities
Administrator	Manage system	Monitor operations and manage users
Municipal Authority	Decision maker	Approve reports and monitor performance

Collection Staff	Waste collection	Empty bins according to assigned routes
Vehicle Driver	Transport waste	Follow optimized collection routes
Citizens	End users	Report overflowing garbage bins
Technician	Maintenance	Maintain sensors and software
Government	Regulatory body	Ensure environmental compliance

4. Existing System Analysis

4.1 Existing Process

The municipality follows a fixed garbage collection schedule. Vehicles travel predetermined routes without checking actual bin status. Workers manually inspect waste bins, resulting in delayed collections and unnecessary trips.

4.2 Strengths

- Easy to operate
- No advanced technology required
- Fixed work schedule
- Simple planning process

4.3 Weaknesses

- Overflowing waste bins
- Fuel wastage
- High operational costs
- Manual monitoring
- Poor route planning
- Delayed collection
- Low efficiency

4.4 Gap Analysis

Current problems not solved include:

- No real-time waste monitoring
- No GPS tracking
- No optimized routes
- No automated notifications
- No data analytics
- Lack of performance monitoring

5. Proposed Software Solution

5.1 Proposed System

The Smart Waste Collection System integrates IoT-enabled waste bins, GPS-enabled vehicles, cloud storage, and a web-based dashboard. Smart sensors measure the waste level inside each bin and send the data to the server in real time. The software identifies bins that require collection, generates optimized collection routes, and helps administrators monitor waste collection activities efficiently.

5.2 Functional Modules

- User Login
- Administrator Dashboard
- Smart Bin Monitoring
- GPS Vehicle Tracking
- Route Optimization
- Driver Assignment
- Notification Module
- Report Generation
- Complaint Management
- System Settings

5.3 Functional Requirements

FR1: The system shall allow administrator login.

FR2: The system shall monitor smart bin fill levels.

FR3: The system shall display bin locations.

FR4: The system shall generate optimized collection routes.

FR5: The system shall assign routes to drivers.

FR6: The system shall send overflow alerts.

FR7: The system shall generate daily reports.

FR8: The system shall maintain collection history.

FR9: The system shall allow citizens to submit complaints.

FR10: The system shall display dashboard analytics.

5.4 Non-Functional Requirements

- High Performance
- Strong Security
- High Availability

- Reliability
- Scalability
- Maintainability
- Usability
- Portability

5.5 Software Engineering Principles Applied

Modularity: Separate modules for monitoring, routing, reporting, and administration.

Scalability: The system can support thousands of smart waste bins across different locations.

Reliability: Continuous monitoring with secure backup storage ensures dependable operation.

Security: User authentication and secure communication protect system data.

Maintainability: The modular architecture allows easy maintenance and future upgrades.

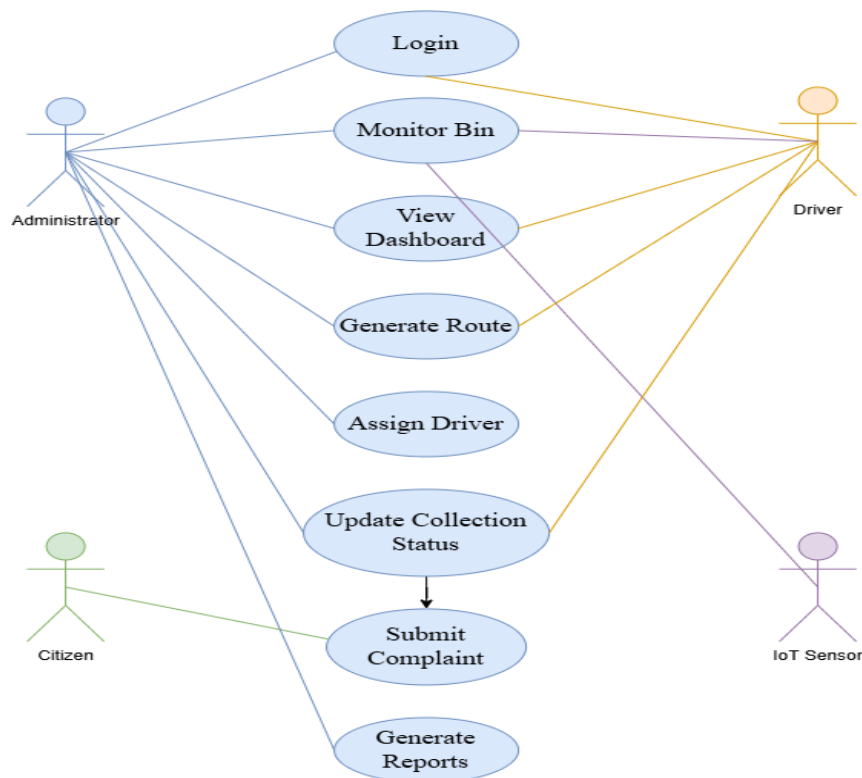
Reusability: Software modules can be reused in other smart city applications.

Usability: A simple dashboard and user-friendly interface make the system easy to use.

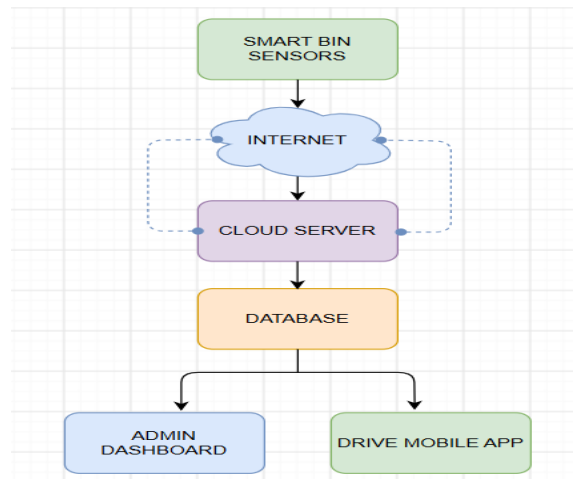
Flexibility: The system supports future enhancements such as AI and advanced analytics.

6. System Design

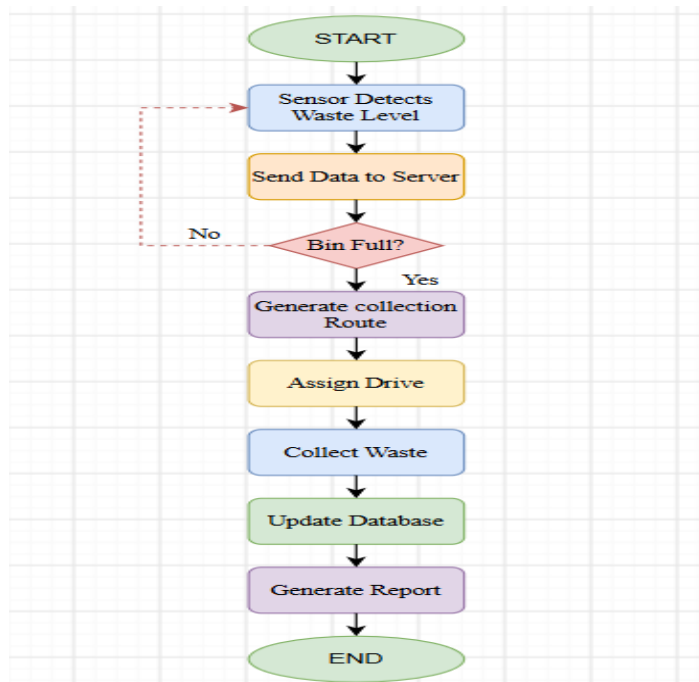
Use Case Diagram



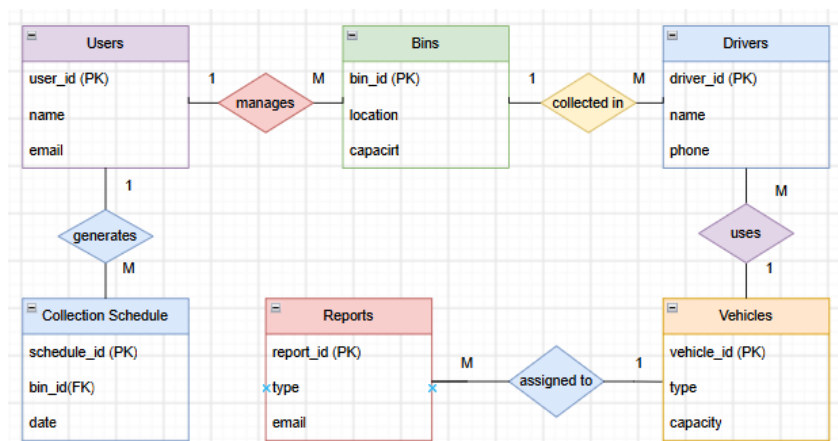
Activity Diagram



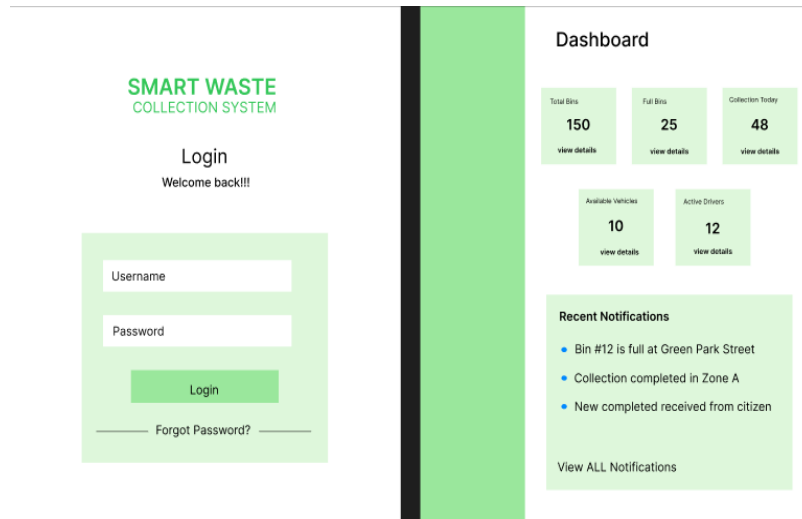
System Architecture



Database ER Diagram



UI/Wireframe:



7. Methodology / SDLC Selection

Selected Model: Agile SDLC

Why Agile?

Agile supports continuous improvement, quick delivery, and easy modification of software based on user feedback. Municipal requirements may change over time, making Agile suitable.

Advantages

- Faster development
- Easy requirement changes
- Continuous testing
- Frequent customer feedback
- Better quality
- Lower project risk

Development Stages

1. Requirement Gathering
2. Sprint Planning
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

8. Implementation Plan

Phase	Duration
Requirement Gathering	1 Week
System Design	1 Week
Development	3 Weeks
Testing	1 Week
Deployment	1 Week
Maintenance	Continuous

9. Expected Outcomes

Benefits

- Cleaner cities
- Reduced operational costs
- Better waste management
- Faster garbage collection
- Reduced fuel consumption

Business Impact

- Lower maintenance cost
- Efficient resource utilization
- Improved productivity

User Benefits

- Better sanitation
- Real-time complaint handling
- Cleaner surroundings

Social Impact

- Improved public health
- Reduced pollution
- Better quality of life

SDG Contribution

Supports SDG 11: Sustainable Cities and Communities by improving urban sanitation and promoting sustainable waste management.

10. Risks and Limitations

- Internet connectivity issues

- Sensor failures
- Data privacy concerns
- Cybersecurity threats
- Budget limitations
- User resistance
- Technical maintenance
- Power failures

11. Future Enhancements

- AI-based waste prediction
- Machine learning for route optimization
- Mobile application
- Cloud-based analytics
- Blockchain for waste tracking
- Smart recycling management
- Drone monitoring
- Voice assistant support

12. Conclusion

The Smart Waste Collection System provides an efficient software solution for modern waste management. By integrating IoT sensors, GPS tracking, and route optimization, municipalities can reduce operational costs, improve cleanliness, and enhance public health. The proposed system supports SDG 11 by promoting sustainable urban development and efficient municipal services.

13. References

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